Decomposing Exchange Rate Responses to News: Evidence from South Africa

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Motivation

- Monetary policy news moves exchange rates, but **how** is less clear for small open economies
- Textbook channel: a tightening raises the domestic interest rate gap, and the currency appreciates
- In practice, the link is weak and unstable: currencies also respond to risk premia and global conditions
- South Africa is a useful case: an open, inflation-targeting economy with a liquid, freely floating currency
- High-frequency identification of policy surprises lets us isolate the news component (Kuttner 2001; Cook and Hahn 1989; Gurkaynak, Sack, and Swanson 2005; Swanson 2021; Miranda-Agrippino and Ricco 2021)

Typical moves on announcement days

Sample	Days	Spot: mean move (%)	Spot: days over 1%	2y gap: mean move (bp)
Other days	3,633	0.82	30%	5.5
SA monetary policy	101	0.88	33%	9.8
SA data releases	424	0.90	35%	7.3
US monetary policy	123	1.13	50%	7.7

- On an ordinary day the rand moves about 0.8%, and on SA monetary policy and data release days it moves about the same (0.88% and 0.90%)
- US monetary policy days stand out: 1.13% on average, with 50% of days moving more than 1%
- The rate gap reacts more clearly on SA monetary policy days: about 10 bp at the 2y, against 6 bp on other days
- The news is a small signal against the rand's everyday volatility, so we need to isolate the surprise

Mean move is the mean absolute daily change. Other days have no announcement.

Related literature: exchange rate responses to news

- Macro announcement surprises cause jumps in exchange rates in high-frequency data (Andersen et al. 2003)
- The joint response of exchange rates and term structures is read through uncovered interest parity and risk premia: a stronger-than-expected release can appreciate the dollar while implying overshooting (Faust et al. 2003)
- Small open economies: an unanticipated 25 bp tightening appreciates the exchange rate by about 0.35% on average, yet policy accounts for only a small share of exchange rate variability (Kearns and Manners 2005; Zettelmeyer 2004)
- New Zealand: data and monetary policy surprises have large effects, and the published rate track adds about 10% explanatory power (Karagedikli and Siklos 2008)

Closest in design to ours

- Coleman and Karagedikli (2012) compare the response of the spot rate, 2y and 5y swap rate differentials and the synthetic forward schedule in New Zealand
- Monetary surprises move spot and the 5y swap rate by similar amounts, so forwards respond little
- CPI and GDP surprises move spot about three times as much as the 5y rate, so forwards appreciate on stronger CPI or GDP news
- This contrasts with standard theory and US evidence
- We ask the same questions for South Africa, with forwards implied by CIP (see appendix)

Related literature: theory, term structure and information effects

- Exchange rates and fundamentals: fundamentals help little in predicting floating exchange rates, which look close to a random walk (Engel and West 2005; Engel et al. 2007)
- Interest rate rules link monetary policy to the real exchange rate (Engel and West 2004)
- What the exchange rate response to inflation news says about monetary policy (Clarida and Waldman 2008)
- Term structure responses to economic and monetary policy surprises (Gürkaynak, Sack, and Swanson 2005)
- Information effects: how forward guidance is worded shapes private sector responses (Lunsford 2020; Jarociński 2024), which motivates the CBI factor

What we add

- Four monetary policy factors for South Africa: Target, FG, CBI and country risk, plus data releases and US surprises
- Responses across the term structure up to 10 years, in spot and the SA–US OIS rate gap
- A split of the rate gap into a news-driven (predictable) and a non-news (unpredictable) part, each related to the rand
- Local projections that show whether a surprise shifts the gap curve in parallel or tilts it

What we ask

1. How do the rand and the SA–US interest rate gap respond to South African monetary policy surprises and to data releases?
2. Does the rand respond to the **news-driven** part of the rate gap, or to the rest?
3. Do US monetary policy surprises work through the same channel?

Data and sample

Data	Event days	Sample
SA monetary policy surprises (Pirozhkova et al. 2024)	101	Apr 2008 – Nov 2024
SA data releases (CPI, GDP, unemployment)	424	Oct 2007 – Dec 2024
US monetary policy surprises (Jarociński 2024)	123	Oct 2007 – Sep 2024
ZARONIA and US OIS rates	Daily	
ZAR/USD spot rate	Daily	

- Regressions use event days only; the SA sample ends in 2024 because of ZARONIA data availability

Four surprise factors

- **Target**: the immediate market reaction to the policy rate decision
- **Forward guidance (FG)**: the response to the communicated longer-term path of policy
- **Central bank information (CBI)**: what the announcement reveals about the central bank's view of the economy (Jarociński 2024)
- **Country risk (CR)**: the central bank's role in responding to external shocks in a small open economy (Pirozhkova, Ricco, and Viegi 2024)
- US surprises are defined in the same way, with an added large-scale asset purchase (LSAP) factor and no CR factor
- Data release surprises: **actual minus the median analyst forecast**

Variables and units

Variable	Definition	Unit
Surprises	Policy factors and release surprises	pp
Spot	Daily log change in ZAR/USD	% (positive = rand depreciates)
OIS gap, tenor n	Change in (SA OIS - US OIS)	pp
Forward, tenor n	Spot change + n x change in OIS gap	%

- Coefficients read as: the response to a **1 pp** surprise
- Tenors: 1, 2, 3, 4, 5 and 10 years

How the rate gap relates to the rand

S is ZAR per USD (a rise is a weaker rand). The gap is the SA minus US OIS rate at tenor n:

$$\text{gap}_t^{(n)} = r_t^{SA,(n)} - r_t^{US,(n)} \tag{1}$$

- **CIP:** the forward premium equals the gap, $\ln F^{(n)} - \ln S \approx n \times \text{gap}^{(n)}$. This ties the gap to the forward, not to spot
- **UIP:** the expected rand depreciation equals the gap. Rearranged, with a risk premium rp_t that is zero under UIP:

$$\ln S_t = E_t[\ln S_{t+n}] - n \times \text{gap}_t^{(n)} + rp_t \tag{2}$$

- A surprise that raises the gap lowers spot today: the rand appreciates by up to $n \times \Delta\text{gap}$ if the long-run rate is unchanged, then drifts weaker (overshooting)
- UIP fails in practice, and a surprise can also shift the long-run rate or the risk premium, so the sign and size of the response are an empirical question (Faust et al. 2003)

Forward rates are implied by the rate gap

Covered interest parity links spot, forwards and the SA–US rate gap:

$$\Delta \ln F_t^{(n)} = \Delta \ln S_t + n\,\Delta \left(r_t^{SA} - r_t^{US}\right) \tag{3}$$

- Our forwards are constructed this way, so they carry no information beyond **spot and the rate gap**
- The focus is therefore on spot and the gap; forwards translate the gap response into exchange-rate terms
- Multiplying by the tenor makes long forwards mechanically more sensitive than short ones

Event-study regressions

$$\Delta y_t = \alpha + \beta\, s_t + \varepsilon_t \tag{4}$$

- y_t: spot or OIS gap at a given tenor (forwards are in the appendix)
- s_t: a monetary policy or data release surprise
- Estimated on announcement or release days
- Newey–West standard errors

Splitting the rate gap into predictable and unpredictable parts

Step 1: regress the change in the gap on the surprise

$$\Delta gap_t = \alpha + \gamma\, s_t + u_t \tag{5}$$

- **Predictable** part: the fitted value (news-driven)
- **Unpredictable** part: the residual

Step 2: relate each part to the exchange rate

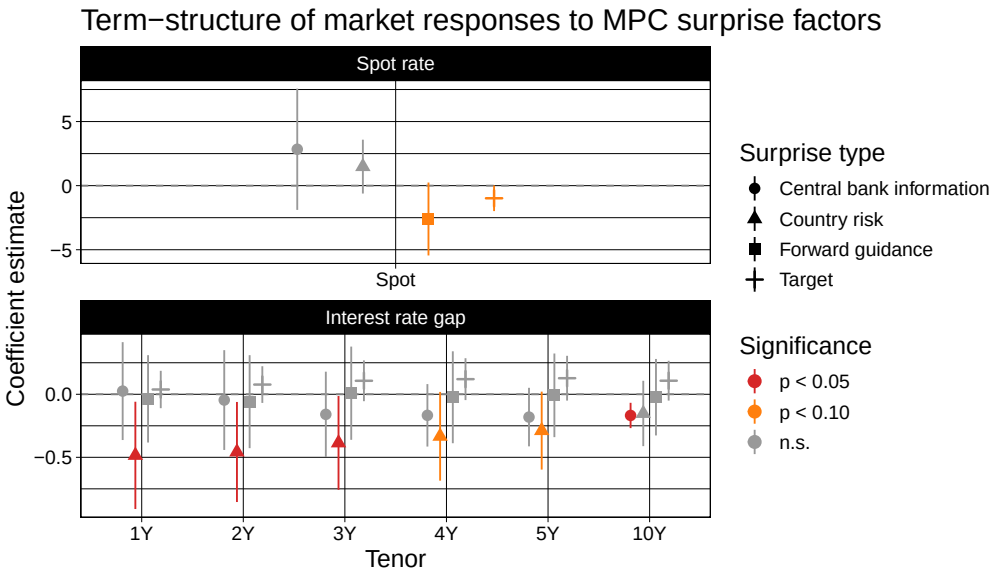
$$\Delta spot_t = a + b_1\, \widehat{gap}_t + b_2\, \hat{u}_t + e_t \tag{6}$$

Baseline: regress spot on the raw change in the gap

Why split the gap into predictable and unpredictable parts?

- The gap moves for many reasons on an announcement day: the policy news itself, other news, risk sentiment and noise
- **Predictable** part: the movement tied to the identified surprise, which is the interest rate channel that theory says should move the currency
- **Unpredictable** part: everything else, including risk premia, information effects and noise. It is not clearly exogenous
- Regressing spot on the raw gap mixes the two, so a real effect can be hidden
- Using the predictable part is equivalent to an instrumental variable estimate with the surprise as the instrument. This assumes the surprise affects spot only through the gap, and it is unreliable when the gap barely responds to the surprise, as for Target and FG
- Comparing the two parts shows whether the rand responds to expected interest rate differentials or to risk premia (Faust et al. 2003)

SA monetary policy surprises: responses across variables



SA monetary policy surprises: significant responses

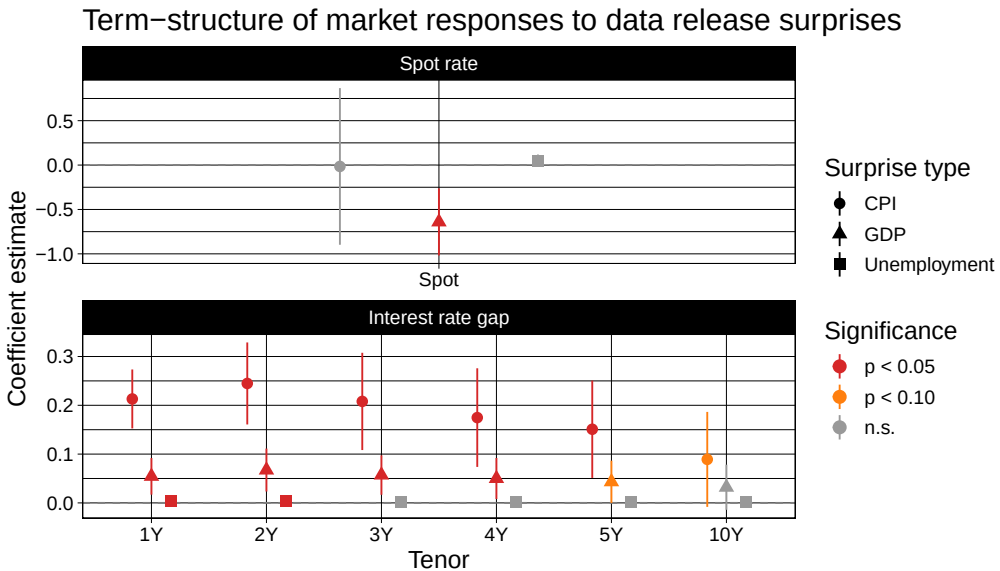
Surprise	Response	Coefficient	Std. error	p-value
Target	Spot	-0.986*	0.511	0.056
Forward Guidance	Spot	-2.603*	1.452	0.076
Central Bank Information	OIS gap 10y	-0.168***	0.051	0.001
Country Risk	OIS gap 1y	-0.485**	0.217	0.028
Country Risk	OIS gap 2y	-0.458**	0.202	0.026
Country Risk	OIS gap 3y	-0.386**	0.190	0.045
Country Risk	OIS gap 4y	-0.333*	0.180	0.066
Country Risk	OIS gap 5y	-0.288*	0.158	0.071

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

SA monetary policy surprises: what stands out

- A positive Target surprise is associated with rand appreciation: spot -0.99 ($p = 0.056$)
- Forward guidance has the larger spot response: -2.60 ($p = 0.076$)
- Both are only borderline significant ($p < 0.10$)
- CBI lowers the long-end rate gap: 10y -0.17 ($p = 0.001$)
- Country risk lowers the short-to-medium gap: 2y -0.46 ($p = 0.026$)
- Most other responses are not distinguishable from zero in a sample of 101 announcement days

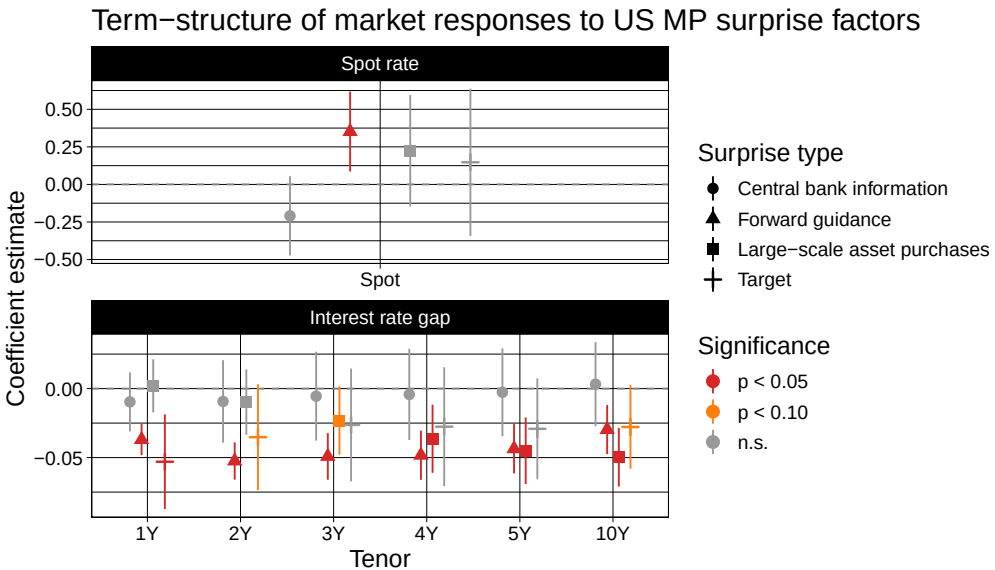
SA data releases



SA data releases: what stands out

- Data releases move the rate gap more clearly than monetary policy surprises do
- A CPI surprise raises the gap at all tenors: 2y 0.24 ($p < 0.001$)
- A GDP surprise strengthens the rand: spot -0.64 ($p = 0.001$), and also raises the short-to-medium gap
- Unemployment has small effects on the gap at short tenors

US monetary policy surprises



US monetary policy surprises: what stands out

- US surprises move the SA–US gap strongly and consistently across tenors
- Federal funds rate surprise: 1y gap -0.05 ($p = 0.003$)
- Forward guidance: 2y gap -0.05 ($p < 0.001$), and the rand depreciates: spot 0.35 ($p = 0.011$)
- LSAP surprises lower the gap at 3y to 10y
- This is the expected direction: a tighter US stance narrows the gap and weakens the rand

Does the rate gap explain the rand? Raw changes

OIS gap	Coefficient	Std. error	p-value	R-squared
10y	2.043	1.349	0.133	0.047
1y	-0.417	1.178	0.724	0.003
2y	-0.268	1.184	0.822	0.001
3y	0.208	1.264	0.870	0.000
4y	1.168	1.332	0.383	0.014
5y	1.515	1.354	0.266	0.024

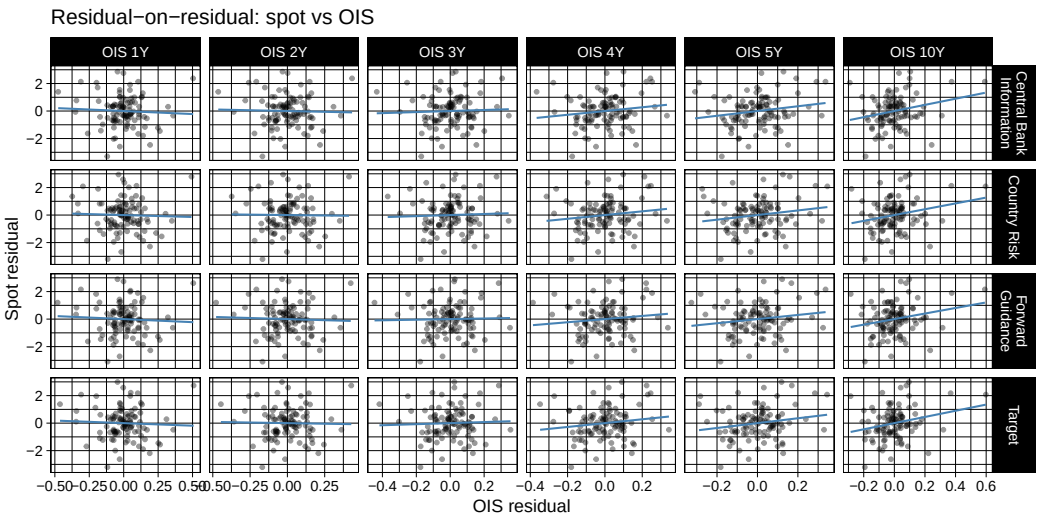
- Raw changes in the gap do not explain spot on announcement days
- R-squared is at most 0.05
- As in the SA surprise results, the shocks that move spot (Target, FG) are not the ones that move the gap (CBI, country risk)

Predictable and unpredictable components

Surprise	OIS gap	Component	Coefficient	Std. error	p-value
Forward Guidance	10y	Predictable	112.135*	58.948	0.060
Forward Guidance	1y	Predictable	71.816*	40.174	0.077
Forward Guidance	2y	Predictable	44.423*	24.485	0.073
Forward Guidance	3y	Predictable	-304.235*	167.630	0.073
Forward Guidance	4y	Predictable	109.657*	59.757	0.070
Forward Guidance	5y	Predictable	313.799*	170.461	0.069
Target	10y	Predictable	-9.167*	4.656	0.052
Target	10y	Unpredictable	2.247*	1.316	0.091
Target	1y	Predictable	-25.968*	13.735	0.062
Target	2y	Predictable	-12.807*	6.882	0.066
Target	3y	Predictable	-9.212*	5.060	0.072
Target	4y	Predictable	-8.219*	4.465	0.069
Target	5y	Predictable	-7.731*	4.174	0.067

- Only combinations significant at 10% are shown
- The **predictable** part is borderline significant for Target and FG because spot responds to the surprise itself. The gap is a weak first stage for these shocks, so this does not show that the gap drives spot

No other common underlying factors

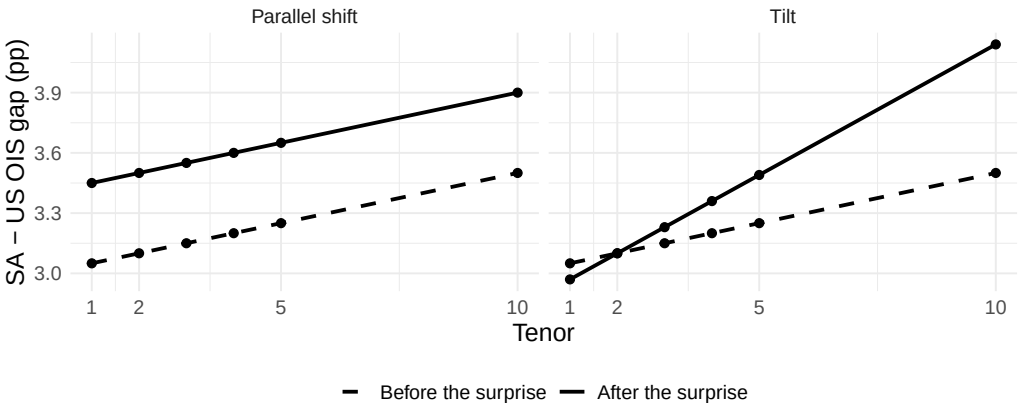


- After removing the surprise from both series, spot and gap residuals do not co-move: 1 of 24 slopes are significant at 10%

Local projections

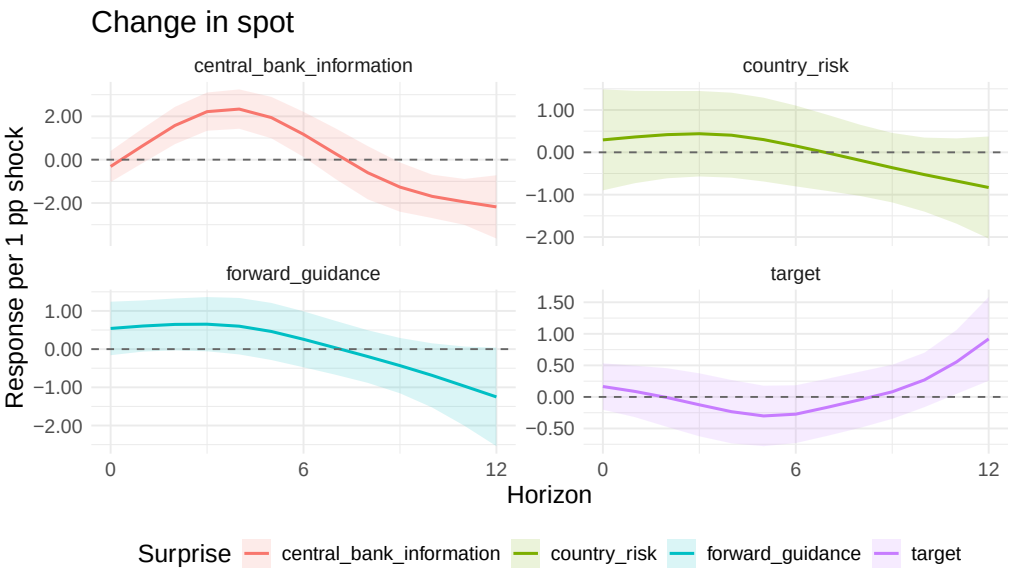
- Smooth local projections of each response on each surprise, one shock at a time
- Responses are scaled to a **1 pp** surprise, with 80% confidence bands
- Horizon is counted in **announcement events**, not days: horizon 0 is the announcement-day response
- Later horizons show how subsequent event-day changes relate to the shock, not the persistence of the shift
- Question: does a surprise shift the rate gap curve in **parallel**, or **tilt** it?
- Level: the 2y OIS gap. Slope: the 10y gap minus the 2y gap

Parallel shift versus tilt (illustration)

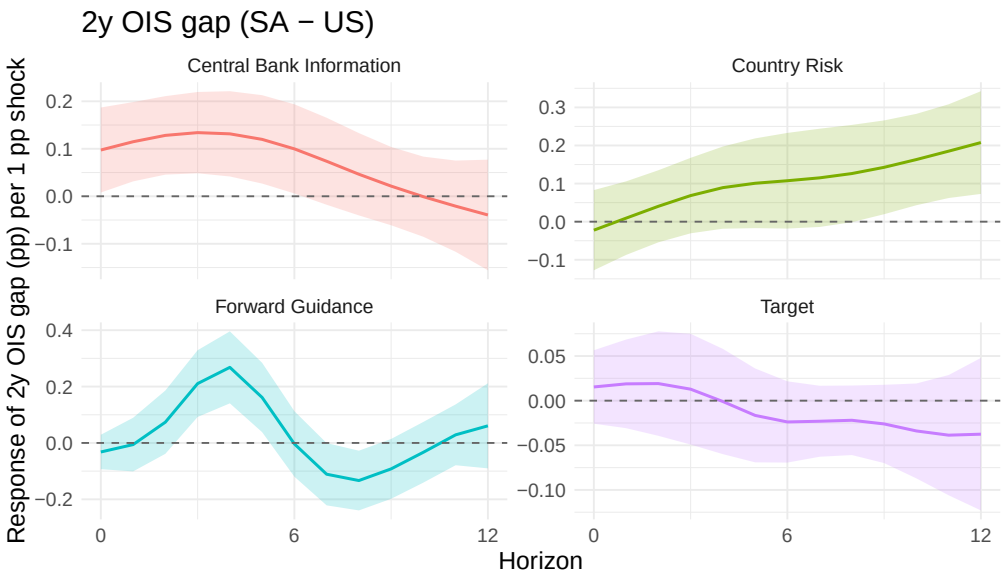


- **Parallel shift:** every tenor moves by the same amount, so the level changes and the slope does not
- **Tilt:** tenors move by different amounts, so the slope changes; here the 2y is the pivot and stays put
- We measure the level as the 2y gap response and the slope as the 10y minus 2y response (illustrative numbers, not estimates)

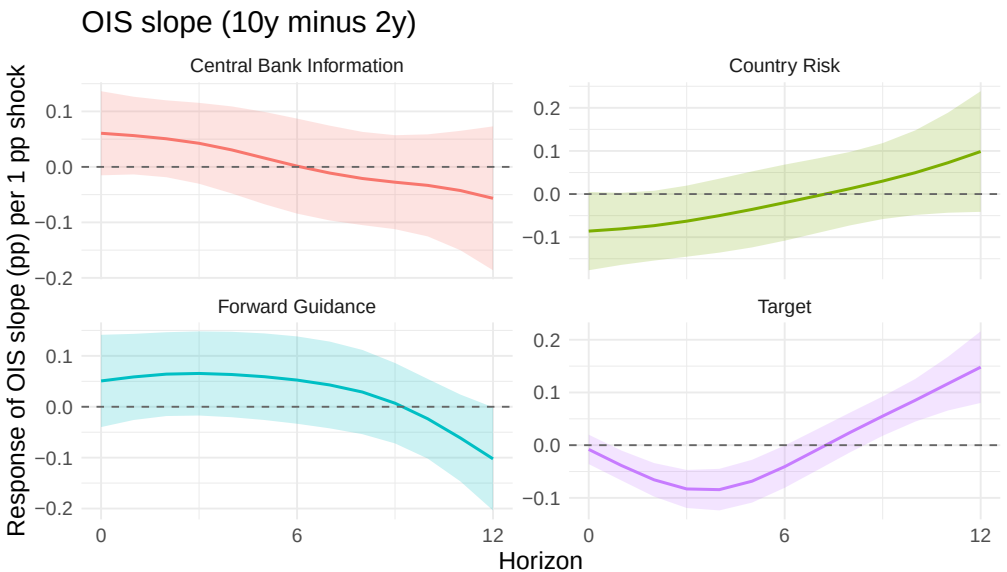
Spot responses by surprise



Level: 2y interest gap



Slope: 10y minus 2y interest gap



Parallel shift or tilt?

- **CBI** and **forward guidance** move the 2y gap with little change in the slope: closer to a parallel shift
- The **target** surprise barely moves the 2y gap but changes the slope: a tilt, with the long end doing the moving
- **Country risk** shows little response until late horizons
- A zero slope response is consistent with a parallel shift, but only if the level moves

Storyline - what are we saying?

1. **SA monetary policy moves the rand, but not through the gap.** Target and FG move spot (-0.99 ($p = 0.056$) and -2.60 ($p = 0.076$)), while the gap responds to CBI and country risk instead
2. **Domestic macro news moves rates, and growth news also moves the rand.** A CPI surprise raises the gap (2y: 0.24 ($p < 0.001$)); a GDP surprise raises the gap and strengthens the rand (-0.64 ($p = 0.001$))
3. **US policy transmits through the gap.** US forward guidance narrows the gap (2y: -0.05 ($p < 0.001$)) and weakens the rand (0.35 ($p = 0.011$))
4. **The gap is a poor day-to-day explanation of the rand.** Raw gap changes explain almost none of the variation in spot (R-squared at most 0.05)

Caveats and next steps

- Small sample of announcement days and borderline significance, with no correction for multiple tests
- Forwards are implied by CIP and add no independent information; observed forwards would allow a **CIP deviation** analysis
- Local projection results can be sensitive to lag length, and horizons are in event time
- Level is proxied by the 2y gap; an average across tenors would separate level and slope more cleanly (debatable)

The End

Thank you

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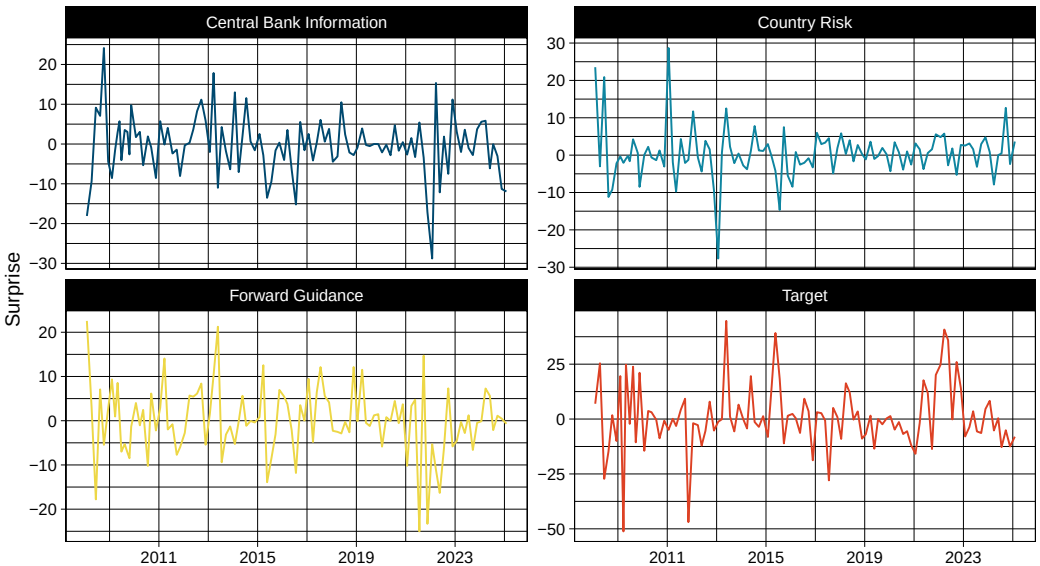
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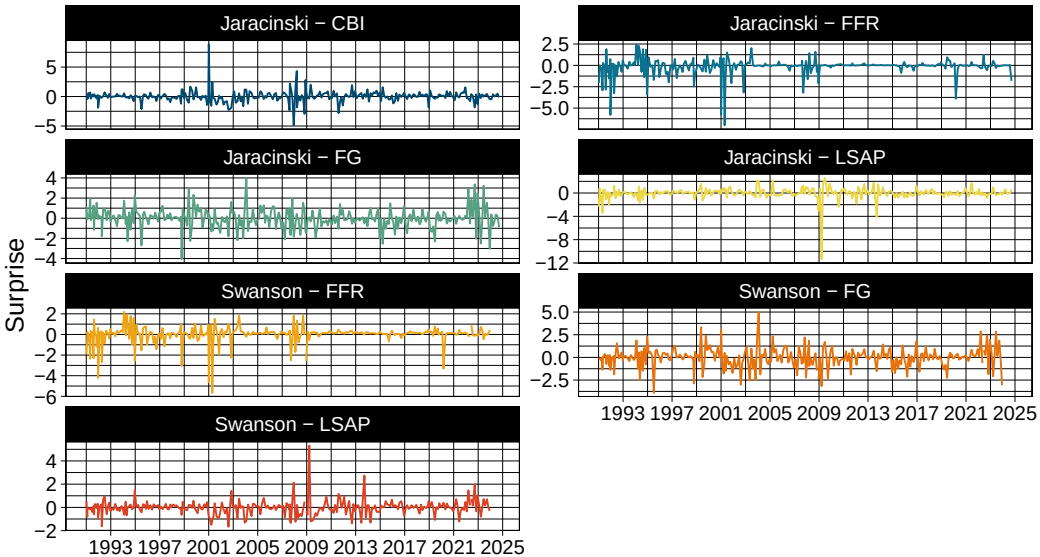
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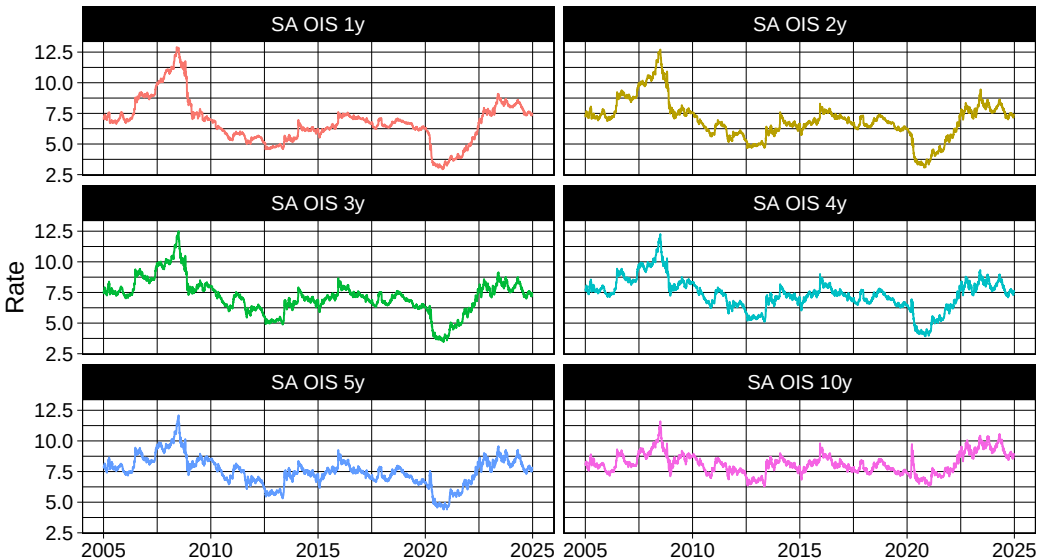
South African monetary policy surprises



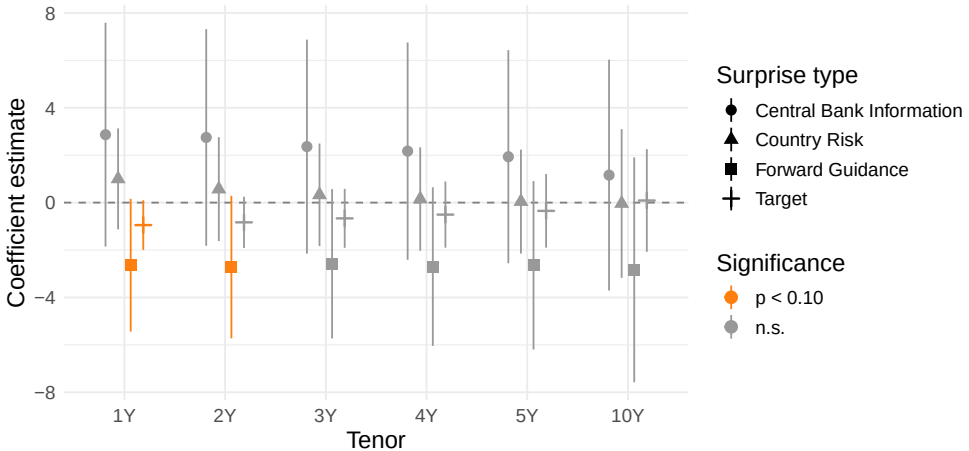
US monetary policy surprises: data



SA OIS rates


SA OIS 1y
SA OIS 2y
SA OIS 3y
SA OIS 4y
SA OIS 5y
SA OIS 10y
Rate
12.5
10.0
7.5
5.0
2.5
2005
2010
2015
2020
2025

Forward responses to SA monetary policy surprises



- Forwards are built as spot plus n times the change in the gap, so they carry no independent information
- At short tenors they mirror the spot response; the gap component grows with the tenor
- Shown for the SA monetary policy surprises only

Descriptive statistics

Series	Mean	Median	SD	Min	Max	Observations
Forward 10y	0.027	-0.039	1.776	-12.036	17.721	4,137
Forward 1y	0.024	-0.023	1.155	-10.665	16.341	4,137
Forward 2y	0.024	-0.023	1.194	-10.993	16.557	4,137
Forward 3y	0.024	-0.021	1.249	-10.956	16.653	4,137
Forward 4y	0.024	-0.012	1.312	-10.983	16.780	4,137
Forward 5y	0.024	-0.020	1.382	-10.775	17.009	4,137
Spot	0.025	0.000	1.137	-10.232	16.213	4,137
OIS gap 10y	0.000	-0.002	0.102	-0.583	0.759	4,137
OIS gap 1y	-0.001	-0.002	0.069	-0.595	0.664	4,137
OIS gap 2y	-0.000	-0.003	0.086	-0.834	0.700	4,137
OIS gap 3y	-0.000	-0.002	0.092	-0.570	0.645	4,137
OIS gap 4y	-0.000	-0.002	0.096	-0.591	0.662	4,137
OIS gap 5y	-0.000	-0.002	0.099	-0.563	0.661	4,137